THE SOLID STATE

Objective Type Questions — Bihar Board Class 12 Chemistry

Total Questions: 100 • Full Marks: 100 • Time: 2 Hours

1. What is the position of particles in solids?

Free | Fixed
Random | None of these

2. Intermolecular forces in solids are-

Weak | Strong | Zero | None

3. Which of the following is an amorphous solid?

Diamond | Graphite | Glass | Salt

4. The arrangement of particles in amorphous solids has-

Long-range order | Short-range order
No order | Complete order

5. Crystalline solids are-

Isotropic | Anisotropic
Both | None of these

6. Which of the following is an amorphous solid?

Graphite (C) | Quartz Glass (SiO_2)
Chrome Alum | Silicon Carbide (SiC)

7. Glass is sometimes called-

Supercooled liquid | Crystalline solid
Gas | Ionic solid

8. An amorphous solid has which of the following properties?

It has long-range order | It has a definite melting point | It is isotropic | It is always ionic

9. A major property of crystalline solids is-

Short-range order | Long-range order
Absence of a definite melting point | Isotropy

10. Ice (H_2O) is what type of molecular solid?

Non-polar molecular solid | Polar molecular solid
Hydrogen-bonded molecular solid | Ionic solid

11. Properties of ionic solids are-

Hard and brittle | High melting point
Poor conductors of electricity | All of the above

12. The presence of free electrons in metallic solids causes-

High electrical conductivity | High thermal conductivity
Lustre | All of the above

13. Metallic bonding is found in-

Metal | Non-metal
Common salt | Water

14. Diamond is what type of crystal?

Ionic crystal | Covalent crystal
Molecular crystal | Metallic crystal

15. During cooling of amorphous solids, generally-

The temperature suddenly becomes constant | A definite melting point is obtained | The temperature continues to decrease gradually | Long-range order develops among the particles

16. Silicon is which of the following?

Fertilizer | Semiconductor | Pesticide | Colloid

17. The number of crystal systems is-

4 | 7 | 14 | 8

18. The number of Bravais lattices in the cubic crystal system is-

3 | 1 | 4 | 14

19. The regular three-dimensional arrangement of particles in a crystal is represented by

Crystal lattice | Molecular bond | Coordination number | Packing efficiency

20. The total number of Bravais lattices is-

7 | 14 | 21 | 28

21. A unit cell is described by how many parameters?

3 | 4 | 6 | 8

22. Particles in a primitive unit cell are present-

Only at the corners | Only at the centre
Only at the faces | None of these

23. Particles in a body-centred unit cell are present-

Only at the corners | At the corners and body centre
Only at the faces | None of these

24. Particles in a face-centred unit cell are present-

Only at the corners | At the corners and body centre
At the corners and face centres | None of these

25. In a cubic system, the values of a, b, c and α, β, γ are-

a≠b≠c, α=β=γ=90°
a=b=c, α=β=γ=90°
a=b≠c, α=β=γ=90°
a=b=c, α=β=γ≠90°

26. In a tetragonal system, the values of a, b, c and α, β, γ are-

a≠b≠c, α=β=γ=90°
a=b=c, α=β=γ=90°
a=b≠c, α=β=γ=90°
a=b=c, α=β=γ≠90°

27. In an orthorhombic system, the values of a, b, c and α, β, γ are-

a≠b≠c, α=β=γ=90°
a=b=c, α=β=γ=90°
a=b≠c, α=β=γ=90°
a=b=c, α=β=γ≠90°

28. In a hexagonal system, the values of a, b, c and α, β, γ are-

a=b≠c, α=β=90°, γ=120°
a=b=c, α=β=γ=90°
a=b≠c, α=β=γ=90°
a≠b≠c, α=β=γ=90°

29. The crystal structure of NaCl is

Hexagonal close packing | Face-centred cubic
Square planar | Body-centred cubic

30. The number of atoms in a face-centred cubic unit cell is-

1 | 2 | 4 | 6

31. The number of atoms in a body-centred cubic unit cell is-

1 | 2 | 4 | 6

32. The number of atoms in a primitive cubic unit cell is-

1 | 2 | 4 | 6

33. The contribution of a corner atom to a unit cell is-

1/8 | 1/4 | 1/2 | 1

34. The contribution of an atom at the face centre to a unit cell is-

1/8 | 1/4 | 1/2 | 1

35. In a face-centred cubic unit cell containing identical atoms, the number of tetrahedral voids is-

8 | 4 | 6 | 12

36. The coordination number in one-dimensional close packing is-

2 | 4 | 6 | 8

37. The coordination number in two-dimensional square close packing is-

2 | 4 | 6 | 8

38. The coordination number in two-dimensional hexagonal close packing is-

2 | 4 | 6 | 8

39. The sequence of layers in hcp structure is-

AAA... | ABAB... | ABCABC... | ABCD...

40. The sequence of layers in ccp structure is-

AAA... | ABAB... | ABCABC... | ABCD...

41. If a metal crystallizes in a hexagonal close-packed (hcp) structure, its coordination number is-

12 | 8 | 4 | 6

42. The coordination number in a simple cubic structure is-

2 | 4 | 6 | 8

43. An octahedral void is surrounded by how many spheres?

4 | 6 | 8 | 12

44. In a close-packed structure, the ratio of the number of tetrahedral voids to octahedral voids is-

1:1 | 2:1 | 1:2 | 4:1

45. A tetrahedral void is surrounded by how many spheres?

4 | 6 | 8 | 12

46. The number of octahedral voids formed by N close-packed spheres is-

N | 2N | N/2 | 4N

47. The number of tetrahedral voids formed by N close-packed spheres is-

N | 2N | N/2 | 4N

48. Which pair of structures has both tetrahedral and octahedral voids?

bcc and fcc | hcp and simple cubic
hcp and ccp | bcc and hcp

49. The total number of voids in one unit cell of ccp structure is-

4 8 12 16

50. The number of octahedral voids per unit cell in ccp structure is-

4 8 12 6

51. The percentage of available empty space in close packing is-

26% 32% 48% 74%

52. The percentage of empty space in a bcc unit cell is-

32% 34% 28% 30%

53. The packing efficiency of hcp and ccp structures is-

52.4% 68% 74% 100%

54. The packing efficiency of bcc structure is-

52.4% 68% 74% 100%

55. The packing efficiency of simple cubic structure is-

52.4% 68% 74% 100%

56. Which type of defect causes a decrease in the density of a crystal?

Frenkel Schottky
Interstitial F-centre

57. In Schottky defect-

Only cations are missing
Only anions are missing
Equal numbers of cations and anions are missing
None of these

58. In Frenkel defect-

A cation leaves its normal site and occupies an interstitial site
An anion leaves its normal site and occupies an interstitial site
Both
None of these

59. Frenkel defect is found in

NaCl KCl ZnS CsCl

60. Schottky defect is found in

ZnS AgCl NaCl AgI

61. F-centre is associated with which defect?

Metal excess defect due to anion vacancies
Frenkel defect
Schottky defect
Impurity defect

62. When an electron is trapped by an anion vacancy in a crystal, which defect is formed?

Schottky defect Frenkel defect
F-centre None of these

63. When NaCl is heated in sodium vapour, the crystal becomes-

Yellow Pink Violet Green

64. When LiCl has excess lithium, the crystal becomes-

Yellow Pink Violet Green

65. When KCl has excess potassium, the crystal becomes-

Yellow Pink Violet Green

66. When ZnO is heated, where are the extra Zn^{2+} ions located in the metal excess defect?

At interstitial sites At normal anion sites
Only at the body centre Only at corner sites

67. Metal deficiency defect is found in

NaCl FeO KCl MgO

68. In FeO with incomplete stoichiometry, the approximate ratio of iron to oxygen is-

1:1 2:1 1:2 0.95:1

69. Due to interstitial defect, the density of the crystal-

Increases Decreases
Remains unchanged None of these

70. Due to vacancy defect, the density of the crystal-

Increases Decreases
Remains unchanged None of these

71. When $SrCl_2$ is added to NaCl as an impurity defect-

Cation vacancies are produced Anion vacancies are produced
No vacancy is produced None of these

72. Which defect is produced on heating a solid?

Schottky Frenkel
Vacancy defect Interstitial defect

73. How many types of imperfections are there in a crystal?

1 2 3 4

74. How many types of point defects are there?

2 3 4 5

75. F-centre imparts to the crystal-

Colour Conductivity
Magnetism None of these

76. Which of the following is a good conductor of electricity?

Si Ge Cu All

77. Which of the following is a semiconductor?

Si Ag Cu Fe

78. The conductivity of electrical conductors (Ω^{-1} m^{-1}) is-

10^4 to 10^7 10^{-6} to 10^4 10^{-20} to 10^{-10} to None

79. The conductivity of semiconductors (Ω^{-1} m^{-1}) is-

10^4 to 10^7 10^{-6} to 10^4 10^{-20} to 10^{-10} to None

80. The conductivity of electrical insulators (Ω^{-1} m^{-1}) is-

10^4 to 10^7 10^{-6} to 10^4 10^{-20} to 10^{-10} to None

81. In an n-type semiconductor, doping is done with an element of-

Group 13 Group 14
Group 15 Group 16

82. In a p-type semiconductor, doping is done with an element of-

Group 13 Group 14
Group 15 Group 16

83. The majority charge carriers in an n-type semiconductor are-

Electrons Holes
Both None of these

84. The majority charge carriers in a p-type semiconductor are-

Electrons Holes
Both None of these

85. The conductivity of semiconductors on increasing temperature-

Increases Decreases
Remains unchanged Becomes zero

86. The substance that exhibits paramagnetism is

H_2O O_2 NaCl C_6H_6

87. Which of the following is a ferromagnetic substance?

NaCl Fe O_2 N_2

88. Paramagnetic substances are-

Weakly attracted Weakly repelled
Strongly attracted None of these

89. Diamagnetic substances are-

Weakly attracted Weakly repelled
Strongly attracted None of these

90. Ferromagnetic substances are-

Weakly attracted Weakly repelled
Strongly attracted None of these

91. An example of antiferromagnetism is-

Fe MnO Fe_3O_4 O_2

92. An example of ferrimagnetism is-

Fe MnO Fe_3O_4 O_2

93. Which of the following oxides shows electrical properties like a metal?

SiO_2 MgO $SO_2(s)$ CrO_2

94. Statement-I: In CsCl crystal, the coordination number of Cs^+ is 8.
Statement-II: In CsCl crystal, Cl^- ions form a simple cubic lattice.

Both statements are correct and Statement-II is the correct explanation of Statement-I.
Both statements are correct, but Statement-II is not the correct explanation of Statement-I.
Statement-I is correct, but Statement-II is false.
Statement-I is false, but Statement-II is correct.

95. What is the coordination number of sodium in sodium oxide (Na_2O)?

6 4 8 2

96. The coordination number of a body-centred atom in a cubic structure is-

4 6 8 9

97. Electrical conduction in metallic solids is mainly due to-

Electrons Ions
Both None of these

98. Heat conduction in metals occurs through-

Electrons Ions
Both None of these

99. Semiconductors are used in-

Diodes Transistors
Solar cells All of the above

100. Ferromagnetic substances have regions called-

Domains Atoms
Molecules None of these